6.2) 1) $\forall n \in N\, \forall A\, \forall R[(|A| = n \wedge R \text{ is partial on } A) \rightarrow \exists T(T \text{ is total on } A \wedge R \subseteq T)]$ — (1)

$|A| = n+1$

R is partial on A → Reflexive, → Antisymmetric, → Transitive — (2)

$A' = A \setminus \{a\}$, $|A'| = n$ — (3)

a is minimal of A under R : $\forall x \in A[xRa \rightarrow x = a]$ — (4)

$A' \subseteq A$ clearly — (5)

(a) Let $R' = R \cap (A' \times A')$ — (6) ⟶ R' is partial on A'

Ref → Supp $x \in A'$. By (5), $x \in A$. As R is partial on A, by (2), $(x,x) \in R$ & by (6) $(x,x) \in R'$.

Antisymmetric → Supp $x \in A', y \in A'$, $xR'y \wedge yR'x$ ⟶ $x = y$

By (5), $x, y \in A$, by (2) $x = y$.

Transitive → Supp $x, y, z \in A'$, $xR'y \wedge yR'z$ ⟶ $xR'z$.

then $x, y, z \in A$ by (3); by (2), xRz & clearly $(x,z) \in A' \times A'$ so by (6), $xR'z$.

(b) $T = T' \cup (\{a\} \times A)$ — (7) ⟶ T is total on A $\wedge$ $R \subseteq T$

where T' is total on A' → Ref, → Antisymm, → Transitive, → $\forall x \in A' \forall y \in A'(xT'y \vee yT'x)$ — (8)

$R' \subseteq T'$ — (9)

To prove $R \subseteq T$.

Supp xRy ⟶ xTy.

Case 1) $x \neq a, y \neq a$. Then by (6), $xR'y$. By (9), $xT'y$. By (7) xTy.

Case 2) $x = a$, $y \neq a$. Then by (7), xTy.

Case 3) $x \neq a$, $y = a$. This contradicts the fact that a is minimal so this case is not possible.

Case 4) $x = a$, $y = a$. Then by ⑦, xTy.

6.2) 3) Base case) Let $|B| = 1$, then $B = \{b\}$. This b is R-smallest & largest element of B.

Inductive step: supp any set $B \subseteq A$ with n elements has smallest & largest element. (I.H.) — ①

Supp $B \subseteq A$ with $|B| = n+1$ & $B' = B \setminus \{b\}$ — ②

so $|B'| = n$ — ③

By ①, $\exists s \rightarrow \forall x \in B' (sRx)$ — ④

$\exists l \rightarrow \forall x \in B' (xRl)$ — ⑤

we can use a line of 3 cases such that b is either smallest, largest or none. But an easier line is to use the fact that R is total order. so sRb or bRs

Case 1) sRb. ⟶ s will be smallest in this case

Supp $x \in B$ be arbitrary.

Case 1.1) $x = b$. Then clearly sRx.

Case 1.2) $x \neq b$. Then by ②, $x \in B'$ & by ④ sRx.

Case 2) bRs ⟶ b will be smallest in this case

supp $x \in B$ be arbitrary

Case 2.1) $x = b$. clearly (xRs or bRs) bRx; or bRb.

Case 2.2) $x \neq b$. Then by ②, $x \in B'$ so sRx. As bRs & sRx. By transitivity of R, bRx.

Similar Reasoning with $bRl \vee lRb$ shows that b or l will be largest element of B. so B will have largest & smallest element in any case. If we go with other line of reasoning considering 3 cases of b being smallest, largest or none. Then

After a long reasoning, we would be able to see which are smallest & largest elements of B in each case but we need not know that. We only need to show that B has smallest & largest element.

6.2) 5) Base case) $F_1 = 2^{2^1}+1 = 5$ & $F_0+2 = (2^{2^0}+1)+2 = (2^1+1)+2 = 5$

Inductive step: Supp for $n \geq 1$, $F_n = (F_0 \cdot F_1 \cdots F_{n-1})+2$ — ①

We have to prove that $F_{n+1} = (F_0 \cdot F_1 \cdots F_{n-1} F_n)+2$ — ②

To use $F_n = 2^{2^n}+1$ — ③. We need to somehow multiply 2^{2^n} with 2^{2^n}. It will give $2^{2^n} \cdot 2^{2^n} = 2^{2^n+2^n} = 2^{2 \cdot 2^n} = 2^{2^{n+1}}$. But directly multiplying ③ by 2^{2^n} will give an extra 2^{2^n} & no +1. The idea is to multiply ③ by $2^{2^n}-1$, so the +ve & -ve 2^{2^n} gets cancelled. So

$$(2^{2^n}+1)(2^{2^n}-1) = 2^{2^{n+1}} - 2^{2^n} + 2^{2^n} - 1$$
$$= 2^{2^{n+1}} - 1.$$

Now we can add 2 to get F_{n+1}. as $2^{2^{n+1}}-1+2 = 2^{2^{n+1}}+1 = F_{n+1}$.

Using this idea, we can multiply both sides of ① by $(2^{2^n}-1)$ & then add 2. This gives F_{n+1} on L.H.S & R.H.S will give as follows.

$$F_{n+1} = (F_0 F_1 \cdots F_{n-1}+2)(2^{2^n}-1)+2$$
$$= F_0 \cdots F_{n-1} \cdot 2^{2^n} - F_0 \cdots F_{n-1} + 2 \cdot 2^{2^n} - 2 + 2$$
$$= F_0 \cdots F_{n-1}(F_n - 1) - (F_n - 2) + 2(F_n - 1)$$

as $F_n = 2^{2^n}+1$ so $F_n - 1 = 2^{2^n}$ by ①

$$= F_0 \cdots F_{n-1} F_n - F_0 \cdots F_{n-1} - F_n + 2 + 2F_n - 2$$
$$= F_0 \cdots F_n - (F_n - 2) + F_n$$
$$= F_0 \cdots F_n - F_n + 2 + F_n$$
$$F_{n+1} = F_0 \cdots F_n + 2 \quad \text{As Required}$$

To make F_{n+1} from F_n, we could also have multiplied by 2^{2^n} & then subtract F_n & add 2 to make $2^{2^{n+1}}+1 = F_{n+1}$

6.2) 7) (a) $(a-b)^2 = a^2 - 2ab + b^2 \geq 0$

Dividing by ab on both sides $\to \frac{a^2}{ab} - \frac{2ab}{ab} + \frac{b^2}{ab} \geq 0$

$\frac{a}{b} - 2 + \frac{b}{a} \geq 0 \to \frac{a}{b} + \frac{b}{a} \geq 2$.

(b) $a, b, c \in \mathbb{R}$ $0 < a \leq b \leq c$

$(c-a)(c-b) \geq 0 \to c^2 - cb - ac + ab \geq 0$

Dividing by ca $\to \frac{b}{a} + \frac{c}{a} - 1 + \frac{b}{c} \geq 0$

$\to \frac{b}{c} + \frac{c}{a} - \frac{b}{a} \geq 1$

(c) Base case) For $n=2$, we need to show for $0 < a_1 \leq a_2$,

$\frac{a_1}{a_2} + \frac{a_2}{a_1} \geq 2$ This follows by part (a). (1)

Inductive step: supp $0 < a_1 \leq a_2 \leq \cdots a_n \to \frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} \geq n$

We have to prove if $0 < a_1 \leq a_2 \leq \cdots \leq a_n \leq a_{n+1}$ are the reals,

then $\frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots \frac{a_{n-1}}{a_n} + \frac{a_n}{a_{n+1}} + \frac{a_{n+1}}{a_1} \geq n+1$ (2)

Adding 1 to both sides of (1),

$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} + 1 \geq n+1$ — (3)

Note by part (b) $\to \frac{a_n}{a_{n+1}} + \frac{a_{n+1}}{a_1} - \frac{a_n}{a_1} \geq 1$ so $\frac{a_n}{a_{n+1}} + \frac{a_{n+1}}{a_1} \geq \frac{a_n}{a_1} + 1$

So we can replace $\frac{a_n}{a_1} + 1$ by $\frac{a_n}{a_{n+1}} + \frac{a_{n+1}}{a_1}$ in (3). This proves the Goal. In other words:

$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots \frac{a_{n-1}}{a_n} + \frac{a_n}{a_{n+1}} + \frac{a_{n+1}}{a_1} \geq \frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} + 1 \geq n+1$

6.2) 12) $|A| = n \to P_2(A) = \frac{n(n-1)}{2}$ — (1)

Prove that $|A| = n+1 \to P_2(A) = \frac{(n+1)n}{2}$ — (2)

Note 2-subsets of $|A| = n$ are $\binom{n}{2} = \frac{n!}{2!(n-2)!} = \frac{n(n-1)}{2}$

Supp $|A| = n+1$. Let $A' = A \setminus \{a\}$ for some $a \in A$ so $|A'| = n$. By (1), $P_2(A') = \frac{n(n-1)}{2}$. There are $\frac{n(n-1)}{2}$ 2-subsets of A that do not contain a (means they are just 2-subsets of A'). The subsets that contain a can be formed by taking any one element from A' among n elements & union it with $\{a\}$. There are n such

2 - subsets so total are:

$$P_2(A) = \frac{n(n-1)}{2} + n = \frac{n^2 - n + 2n}{2} = \frac{n^2+n}{2} = \frac{n(n+1)}{2} \text{ as Req.}$$

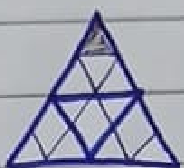
6.2) 13) Base Case: An equilateral △ cut into $4^1 = 4$ △s with one △ removed has 3 △s left which can be placed by a trapezoidal. e.g.

 or 

Inductive step: suppose the statement holds for $n \geq 1$. We have to prove that an equilateral △ divided into 4^{n+1} equilateral △s can be placed with trapezoidals with its one △ removed.

Since an equilateral △ is divided into equilateral △s, it has line segments dividing its left & right sides into 2 equal lengths. We can form an inverted △ in center from the line segment horizontally touching 2 sides of △ at center.

Example for $n=2$.



This way, we can divide △ into 4 equilateral △s of 4^n smaller Triangles inside each as total $4^n + 4^n + 4^n + 4^n = 4 \cdot 4^n = 4^{n+1}$. When any of these △ has 1 corner removed, it can be placed by trapezoidals by inductive hypothesis. For remaing 3 △s, place a trapezoidal at center of opposite edge where a △ was removed. This covers one △ from each of 3 equilateral △s leaving them one corner removed. By inductive hypothesis, each of these can now be placed with ⏢s.

↳ Trapezoidal placed making 1 corner removed from each of 3 △s.